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Fanconi Syndrome and Other Proximal Tubule Disorders

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The proximal tubule reabsorbs the majority of several key solutes, including glucose, amino acids, bicarbonate, and phosphate. This chapter describes a number of disorders, mainly heritable, that affect proximal tubule reabsorption. Chapters 11 and 13 discuss familial forms of hyperphosphaturia and renal tubular acidosis (RTA), respectively.

Most nonelectrolyte solutes are reabsorbed in the proximal tubule through specific transport proteins that cotransport them in conjunction with sodium (Fig. 50.1). The driving force for this solute transport is the electrochemical gradient for sodium entry maintained by the enzyme Na^+,K^+-ATPase. Most disorders of isolated solute reabsorption are related to defects in specific transport proteins, whereas disorders affecting multiple solutes, such as Fanconi syndrome, are probably secondary to defects in energy generation, Na^+,K^+-ATPase activity, or dysfunction of cellular organelles involved with membrane protein recycling.

FANCONI SYNDROME

Definition

In the 1930s, de Toni, Debré, and coworkers and Fanconi independently described several children with the combination of renal rickets, glycosuria, and hypophosphatemia. Fanconi syndrome now refers to global dysfunction of the proximal tubule leading to excessive urinary excretion of amino acids, glucose, phosphate, bicarbonate, uric acid, and other solutes handled by this nephron segment. These losses lead to the clinical problems of acidosis, dehydration, electrolyte imbalance, rickets, osteomalacia, and growth failure. Numerous inherited or acquired disorders are associated with Fanconi syndrome (Box 50.1).

Etiology and Pathogenesis

The sequence of events leading to Fanconi syndrome is incompletely defined and probably varies with each cause. Possible mechanisms include widespread abnormality of most or all of the proximal tubule carriers, such as a defect in sodium binding to the carrier or insertion of the carrier into the brush border membrane, "leaky" brush border membrane or tight junction, inhibited or abnormal Na^+,K^+-ATPase pump, or impaired mitochondrial energy generation (see Fig. 50.1). An abnormality in energy generation has been implicated in multiple disorders, including hereditary fructose intolerance, galactosemia, mitochondrial cytopathies, and heavy metal poisoning, as well as in several experimental models of Fanconi syndrome. Abnormal subcellular organelle function, such as the lysosome in cystinosis or the megalin-cubilin endocytic pathway in Dent disease, is also a cause of Fanconi syndrome (Fig. 50.2).

In adults, the most common causes of persistent Fanconi syndrome are an endogenous or exogenous toxin such as a heavy metal, a medication, or dysproteinemia. In children, the most common persistent cause is an inborn error of metabolism, such as cystinosis. Specific causes of Fanconi syndrome are discussed after a general description of the clinical manifestations and treatment of the syndrome.

Clinical Manifestations

Fanconi syndrome gives rise to a number of clinical abnormalities (Box 50.2).

Aminoaciduria

Aminoaciduria is a cardinal feature of Fanconi syndrome. Virtually every amino acid is found in excess in the urine, thus the term *generalized aminoaciduria*. However, there are no clinical consequences because the losses are trivial (0.5–1 g/day) in relation to dietary intake.

Glycosuria

Glycosuria in the absence of hyperglycemia is another of the cardinal features of Fanconi syndrome and results from impaired tubular reabsorption of glucose. It is often one of the first diagnostic clues (Fig. 50.3). As with aminoaciduria, glycosuria rarely causes symptoms such as weight loss or hypoglycemia.

Hypophosphatemia

Hypophosphatemia from impaired phosphate reabsorption is common in Fanconi syndrome. Tubular phosphate handling can be assessed by measuring the tubular reabsorption of phosphate (TRP), which is normally greater than 80%, by the following equation:

$$\text{TRP} = 1 - (U_p \times P_c / U_c \times P_p) \times 100\,\%$$

where U_p, P_p, U_c, and P_c are the urinary and serum phosphate and creatinine concentrations, respectively. Subtle changes in phosphate reabsorption can be assessed by measuring the maximum phosphate reabsorption in relation to the glomerular filtration rate (TmP/GFR) on fasting urine and blood samples, as shown in Fig. 11.15. Elevated parathyroid hormone (PTH) and low vitamin D levels also may play a role in the phosphaturia of Fanconi syndrome, although these hormonal abnormalities are not always present. A few patients have impaired conversion of 25-hydroxyvitamin D to 1,25-hydroxyvitamin D; metabolic acidosis, another feature of Fanconi syndrome, also may impair this conversion. Another mechanism for the hypophosphatemia is impairment of the megalin-dependent reabsorption and degradation of filtered PTH.[1] Unabsorbed PTH then binds to receptors in more distal portions of the proximal tubule, leading to increased endocytosis of apical phosphate transporters and increased phosphaturia. The hypophosphatemia, especially if accompanied by

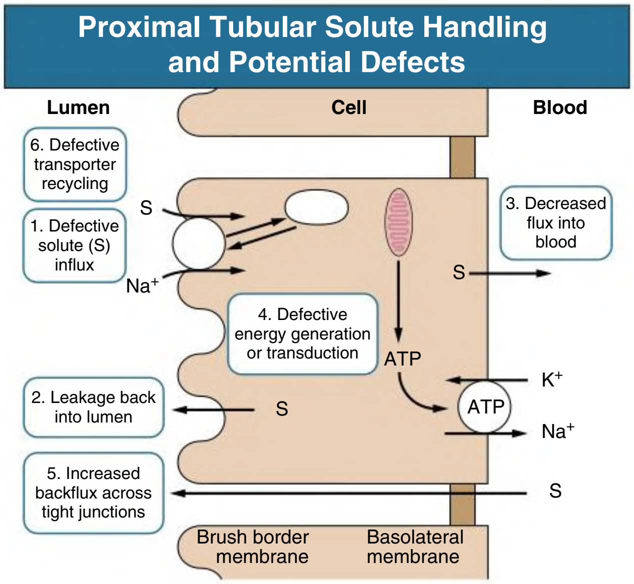

Fig. 50.1 Defects and Potential Defects in Proximal Tubular Solute Handling. Solute uptake by the brush border membrane from the lumen is coupled to Na^+ influx. The favorable electrochemical driving force for luminal Na^+ is maintained by the Na^+,K^+-ATPase pump. Transported solute is then either used by the cell or returned to the blood across the basolateral membrane. Fanconi syndrome could arise because of a defect in one of six areas as shown. *ATP,* Adenosine triphosphate.

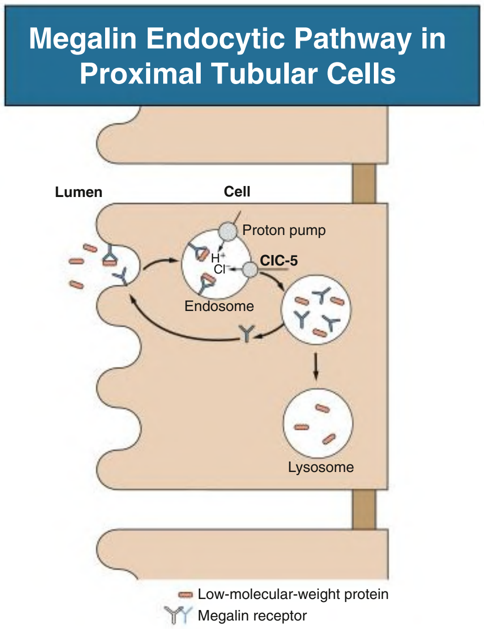

Fig. 50.2 Megalin-Cubilin Endocytic Pathway in Proximal Tubular Cells. Low-molecular-weight proteins in the luminal fluid bind to the megalin-cubilin complex and are endocytosed. The recycling of megalin and further catabolism of these proteins depend on acidification of the vesicle by a proton pump. The ClC-5 chloride channel provides an electrical shunt for efficient functioning of the proton pump. This endocytosis pathway plays a role in membrane transporter recycling, and disruption of this pathway interferes with absorption of other luminal solutes.

BOX 50.1 Causes of Fanconi Syndrome

Inherited

- Cystinosis
- Galactosemia
- Hereditary fructose intolerance
- Tyrosinemia
- Wilson disease
- Lowe syndrome
- Dent disease
- Glycogenosis
- Mitochondrial cytopathies
- Idiopathic

Acquired[a]

- Drugs: *cisplatin, ifosfamide, tenofovir, cidofovir, adefovir,* didanosine, gentamicin, azathioprine, valproic acid (sodium valproate), suramin, streptozocin (streptozotocin), ranitidine
- Dysproteinemias: multiple myeloma, Sjögren syndrome, light-chain proteinuria, amyloidosis
- Heavy metal poisoning: lead, cadmium
- Other poisonings: Chinese herbal medicine, glue sniffing
- Other: nephrotic syndrome, kidney transplantation, acute tubular necrosis

[a]Italics indicate more common causes.

BOX 50.2 Features of Fanconi Syndrome

Metabolic Abnormalities

- Glycosuria
- Hyperaminoaciduria
- Hypophosphatemia
- Acidosis
- Hypokalemia
- Hypouricemia
- Hypocarnitinemia

Clinical Features

- Rickets, osteomalacia
- Growth retardation
- Polyuria
- Dehydration
- Proteinuria

hyperparathyroidism and low 1,25-hydroxyvitamin D levels, often leads to significant bone disease, manifesting with pain, fractures, rickets, or growth failure.

Natriuresis and Kaliuresis

Natriuresis and kaliuresis are common in Fanconi syndrome and can give rise to significant and even life-threatening problems. The decreased proximal reabsorption of sodium leads to increased potassium excretion secondary to increased distal delivery of sodium and activation of the renin-aldosterone system from hypovolemia. In some cases, sodium and potassium losses are so great that hypokalemia and metabolic alkalosis result, simulating Bartter syndrome despite the lowered bicarbonate threshold.

Hyperchloremic Metabolic Acidosis

Hyperchloremic metabolic acidosis, another feature of Fanconi syndrome, is a result of impaired bicarbonate reabsorption by the proximal tubule (proximal or type 2 RTA; see Chapter 13). This impaired reabsorption can lead to the loss of more than 30% of the normal filtered load of bicarbonate. As the serum bicarbonate concentration ($[HCO_3^-]$) falls, the filtered load falls, and excretion drops such that the serum $[HCO_3^-]$ usually remains between 12 and 18 mmol/L. On occasion, there is an associated defect in distal acidification, usually

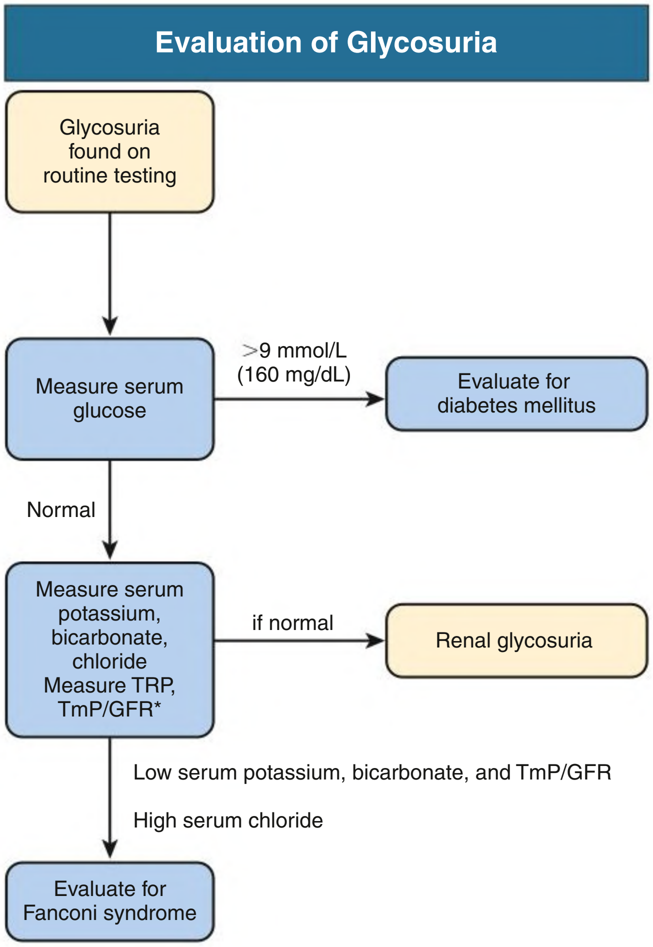

Fig. 50.3 **Evaluation of Glycosuria.** *GFR,* Glomerular filtration rate; *TmP,* tubular maximum reabsorptive capacity for phosphate; *TRP,* tubular reabsorption of phosphate.

in association with long-standing hypokalemia or nephrocalcinosis. Ammoniagenesis is usually normal or increased because of the hypokalemia and acidosis unless there is an associated impairment in glomerular filtration rate (GFR).

Polyuria and Polydipsia

Polyuria, polydipsia, and frequent bouts of severe dehydration are common symptoms in young patients with Fanconi syndrome. The polyuria is mainly related to the osmotic diuresis from the excessive urinary solute losses; but some patients have an associated concentrating defect, especially those with prolonged hypokalemia.

Growth Retardation

Growth retardation in children with Fanconi syndrome is multifactorial. Hypophosphatemia, disordered vitamin D metabolism, and acidosis contribute to growth failure, as do chronic hypokalemia and extracellular volume contraction. Glycosuria and aminoaciduria probably do not play a role. However, even with correction of all these metabolic abnormalities, most patients fail to grow, especially those with cystinosis.

Hypouricemia

Hypouricemia, caused by impairment in the kidney's handling of uric acid, is often present in Fanconi syndrome, especially in adults. Urolithiasis from the uricosuria only rarely has been reported, probably because the urine flow and pH are increased, inhibiting uric acid crystallization.

Proteinuria

Proteinuria is usually minimal, except when Fanconi syndrome develops in association with the nephrotic syndrome. Typically, only low-molecular-weight proteins (<30,000 Da) are excreted, such as vitamin D and A binding proteins, enzymes, immunoglobulin light chains, and hormones. Therefore, significant albuminuria is often absent and urine dipstick may be negative for protein.

Treatment of Fanconi Syndrome

Therapy should be directed at the underlying causes of Fanconi syndrome when possible (see later discussion). This includes avoidance of the offending nutrient in galactosemia, hereditary fructose intolerance, or tyrosinemia; penicillamine or other copper chelators for Wilson disease; or chelation therapy for heavy metal intoxication. In these patients, resolution of Fanconi syndrome usually is complete.

In all patients with Fanconi syndrome, therapy is also directed at the biochemical abnormalities secondary to the solute losses and the bone disease often present in these patients. The proximal RTA (type 2 RTA) usually requires large doses of alkali for correction. Some patients benefit from hydrochlorothiazide to minimize the volume expansion associated with these large doses of alkali. Potassium supplementation usually is also needed, especially if there is significant RTA. If given in combination with a metabolizable anion, such as potassium citrate, lactate, or acetate, these supplements will also correct the acidosis. A few patients will require sodium supplementation along with potassium, and even fewer will require sodium chloride supplementation (especially those who have alkalosis as a result of volume contraction from large urinary NaCl losses). Magnesium supplementation may be required to correct the hypokalemia, and adequate fluid intake is essential. Correction of hypokalemia and its effect on the concentrating ability of the distal tubule may lessen the polyuria.

Bone disease is multifactorial, including urinary loss of vitamin D binding protein and vitamin D, decreased synthesis of calcitriol in some patients, hypercalciuria, chronic acidosis, and hypophosphatemia, which is the major factor. Hypophosphatemia should be treated with 1 to 3 g/day of oral phosphate with the goal of normalizing serum phosphate levels. Many patients with Fanconi syndrome will require supplemental vitamin D for adequate treatment of the rickets and osteomalacia. It is unclear whether standard vitamin D (calciferol [ergocalciferol]) or a vitamin D metabolite is better for supplementation, but most clinicians use a vitamin D metabolite, such as 1,25-dihydroxycholecalciferol (calcitriol). These metabolites obviate the concern of inadequate vitamin D hydroxylation by the proximal tubule mitochondria and reduce the risk for prolonged hypercalcemia because of their shorter half-life. Vitamin D therapy will also improve the hypophosphatemia and lessen the risk for hyperparathyroidism. Supplemental calcium is indicated in those with hypocalcemia after supplemental vitamin D is started. Hyperaminoaciduria, glycosuria, proteinuria, and hyperuricosuria usually do not require specific treatment. Carnitine supplementation, to compensate for the urinary losses, may improve muscle function and lipid profiles, but the evidence is inconsistent.

INHERITED CAUSES OF FANCONI SYNDROME

Cystinosis

Definition

Cystinosis, or cystine storage disease, is characterized biochemically by excessive intracellular storage, particularly in lysosomes, of the amino acid cystine.[2] Three different types of cystinosis can be distinguished based on clinical course, age at onset, and intracellular cystine content. Benign or adult cystinosis is associated with cystine crystals in the cornea and bone marrow only, as well as the mildest elevation in intracellular cystine levels; no kidney disease occurs. Infantile or nephropathic cystinosis is the most common form of cystinosis and is associated with the highest intracellular levels of cystine and the earliest onset of kidney disease. In the intermediate or adolescent form, intracellular cystine levels are between those of the infantile and adult forms, with later onset of kidney disease.

Etiology and Pathogenesis

Nephropathic cystinosis is transmitted as an autosomal recessive trait localized to the short arm of chromosome 17, with an estimated incidence of 1 in 200,000 live births. The *CTNS* gene codes for a lysosomal membrane protein, cystinosin, that mediates the transport of cystine from the lysosome.[3] The benign and intermediate forms of cystinosis are also associated with *CTNS* mutations but still have some functional transport protein, leading to lower intracellular cystine levels and slower onset of kidney disease in the intermediate form and no kidney disease in the benign form. Cystinosin has been shown to play a role in other cellular processes besides lysosomal cystine transport, which may explain why Fanconi syndrome sometimes persists despite cystine depletion.[4]

Clinical Manifestations

The first clinical evidence of nephropathic cystinosis relates to Fanconi syndrome and usually appears in the second half of the first year of life. Subtle abnormalities of tubular function can be demonstrated earlier in families with index cases, but there always is a delay between birth and the first symptoms. Rickets is common after the first year of life, along with growth failure. The growth failure occurs before the GFR declines and despite correction of electrolyte and mineral deficiencies. The GFR invariably declines, and end-stage kidney disease (ESKD) occurs by late childhood.

Nephrocalcinosis is relatively common, and a few patients have kidney stones. Photophobia is another common symptom that occurs by 3 years of age and is progressive. Older patients with cystinosis may develop visual impairment and blindness. Children with cystinosis usually have a fair complexion and blond hair, but dark hair has been observed in some. Cystinosis has been observed in other ethnic groups but is less common than in Whites.

The diagnosis is based on the demonstration of elevated intracellular levels of cystine, usually in white blood cells or skin fibroblasts. Patients with nephropathic and intermediate cystinosis have intracellular cystine levels that exceed 2 nmol half-cystine per milligram of protein (normal <0.2 nmol half-cystine per milligram of protein). Heterozygotes for cystinosis have levels that range from 0.2 to 1 nmol half-cystine per milligram of protein. A slit-lamp demonstration of corneal crystals strongly suggests the diagnosis[2] (Fig. 50.4). A prenatal diagnosis can be made with amniocytes or chorionic villi.

Common late complications of cystinosis include hypothyroidism, splenomegaly and hepatomegaly, decreased visual acuity, swallowing difficulties, pulmonary insufficiency, and corneal ulcerations.[5] Less frequently, older patients have insulin-dependent diabetes mellitus, myopathy, and progressive neurologic disorders. Decreased brain cortex also has been noted on imaging in some patients. Older patients may develop vascular calcification, especially of the coronary arteries, which can lead to myocardial ischemia.

Kidney Pathology

The morphologic features of the kidney in cystinosis vary with the stage. Early in the disease, cystine crystals are present in tubular epithelial cells, interstitial cells, and rarely glomerular epithelial cells[6,7] (Fig. 50.5A). A swan-neck deformity or thinning of the first part of the proximal tubule is an early finding but is not unique to cystinosis. Later, there is pronounced tubular atrophy, interstitial fibrosis, and abundant crystal deposition with giant cell formation of the glomerular visceral epithelium, segmental sclerosis, and eventual glomerular obsolescence. Electron microscopy (EM) demonstrates intracellular crystalline inclusions consistent with cystine (see Fig. 50.5B). Peculiar "dark cells," unique to the cystinotic kidney, also have been observed. These are mostly macrophages and some podocytes and are probably dark because of a reaction of cystine with osmium tetroxide.

Fig. 50.4 Corneal Opacities in Cystinosis. Tinsel-like refractile opacities in the cornea of a patient with cystinosis under slit-lamp examination. (From Foreman JW. Cystinosis and the Fanconi syndrome. In: Avner ED, Harmon WE, Niaudet P, eds. *Pediatric Nephrology*. 5th ed. Lippincott Williams & Wilkins; 2004.)

Treatment

Nonspecific therapy for infantile cystinosis consists of vitamin D therapy and replacement of the urinary electrolyte losses, followed, in due course, by the management of the progressive kidney disease (Table 50.1). Cysteamine therapy can lower tissue cystine levels and slow the decline in GFR, especially in children with normal serum creatinine concentration who are treated before 2 years of age[8] (Fig. 50.6). Cysteamine therapy also improves linear growth but not Fanconi syndrome. The most common problems associated with cysteamine therapy are nausea, vomiting, and a foul odor and taste. Treatment should begin with a low dose of cysteamine soon after the diagnosis is made, increased during 4 to 6 weeks to 60 to 90 mg/kg/day in four divided doses as close to every 6 hours as possible. Slowly increasing the dose minimizes the risk for neutropenia, rash, and arthritis. Leukocyte cystine levels should be checked every 3 to 4 months to monitor effectiveness and compliance, with the goal of achieving and maintaining a cystine level less than 2.0 and preferably 1.0 nmol half-cystine/mg protein. A long-acting formulation of cysteamine is available that allows twice-daily dosing. This formulation is equally effective and may aid compliance, but is significantly more expensive than the standard formulation of cysteamine. Two preparations of topical cysteamine for treatment of the corneal crystals are available: an aqueous solution that is given every hour and a gel formulation that is applied four times a day. Studies are underway to assess the utility of a gene-edited autologous bone marrow transplant in treating cystinosis.

Treatment of ESKD in these children poses no greater problems than in other children. Successful kidney transplantation reverses the kidney failure and Fanconi syndrome but does not appear to improve the extrarenal manifestations of cystinosis. Cysteamine therapy should be continued after transplantation. Cystine does not accumulate in the transplanted kidney, except in infiltrating immunocytes.

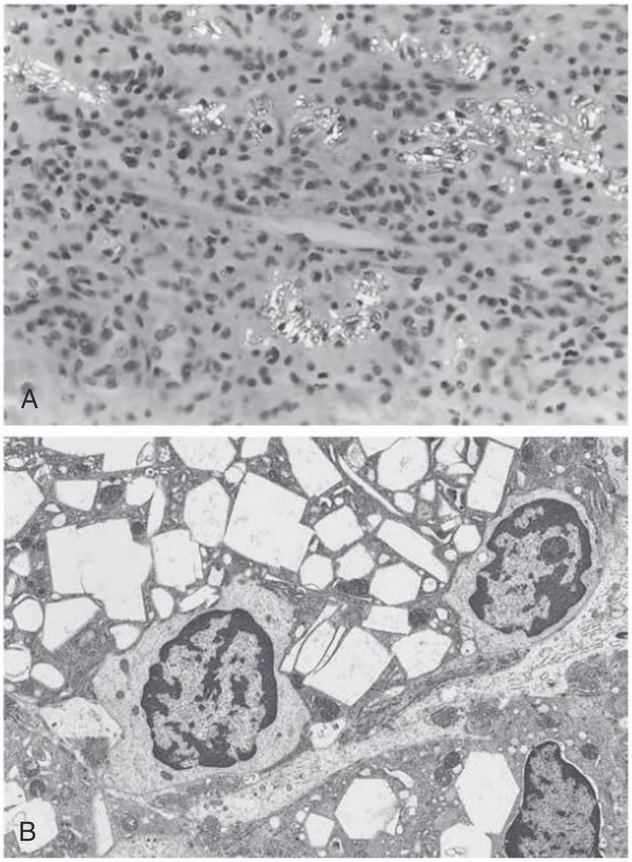

Fig. 50.5 **Cystine Crystals in the Kidney in Cystinosis.** (A) Crystals are seen in photomicrograph of alcohol-fixed nephrectomy specimen, taken through incompletely crossed polarizing filters. Birefringent crystals are evident in tubular epithelial cells and free in the interstitium. (B) Electron micrograph of a renal biopsy specimen shows hexagonal, rectangular, and needle-shaped crystals in macrophages within the interstitium. (Original magnification ×3000.) (A, From Schnaper HW, Cottel J, Merrill S, et al. Early occurrence of end-stage renal disease in a patient with infantile nephropathic cystinosis. *J Pediatr.* 1992;120:575–578. B, From Van't Hoff WG, Ledermann SE, Waldron M, Trompter RS. Early-onset chronic renal failure as a presentation of infantile cystinosis. *Pediatr Nephrol.* 1995;9:483–484.)

TABLE 50.1 Treatment of Cystinosis

Problem	Therapy
Removal of lysosomal cystine	Cysteamine, 0.325 g/m^2 q6h Delayed-release cysteamine, 0.65 g/m^2 q12h Goal: Maintain leukocyte cystine level <1 nmol half-cystine[a]/mg protein
Correction of Tubulopathy	
Dehydration	2–6 L/day fluid
Acidosis	2–15 mmol/kg/day K^+ citrate
Hypophosphatemia	1–4 g/day K^+ phosphate
Rickets	0.25–1 μg/day calcitriol
Adjunct therapies	NaCl, carnitine, indomethacin, hydrochlorothiazide
Later Therapies	
Growth failure	Growth hormone
Hypothyroidism	Thyroxine
Kidney failure	Renal replacement therapy, ideally kidney transplantation

[a]By convention, units are half-cystine because the cystine originally was converted to two cysteine molecules, or "broken in half," before measurement.

Galactosemia

Etiology and Pathogenesis

Galactosemia is an autosomal recessive disorder of galactose metabolism. It is most often the result of deficient activity of the enzyme galactose-1-phosphate uridyltransferase, the incidence of which is 1 in 62,000 live births.[9] Deficiency of this enzyme leads to the intracellular accumulation of galactose-1-phosphate, with damage to the liver, proximal renal tubule, ovary, brain, and lens. A less frequent cause of galactosemia is a deficiency of galactose kinase, which forms galactose-1-phosphate from galactose. Cataracts are the only manifestation of this form of galactosemia.

The pathogenesis of the symptoms of galactosemia is not clear. Accumulation of galactose-1-phosphate subsequent to the ingestion of galactose can inhibit pathways for carbohydrate metabolism, and its level correlates somewhat with clinical symptoms. Defective galactosylation of proteins also has been postulated. Formation of galactitol from galactose by aldose reductase is probably responsible for the cataract formation.

Clinical Manifestations

Affected infants ingesting milk containing lactose (the most common source of galactose in the diet) rapidly develop vomiting, diarrhea, and failure to thrive. Jaundice from unconjugated hyperbilirubinemia is common, along with severe hemolysis. Continued intake of galactose leads to hepatomegaly and cirrhosis. Cataracts appear within days after birth, although at first they often are detectable only with a slit lamp. Cognitive dysfunction or developmental delay may develop within a few months. Fulminant *Escherichia coli* sepsis has been described, possibly a consequence of inhibited leukocyte bactericidal activity.

In addition to these clinical findings, galactose intake leads within days to hyperaminoaciduria and albuminuria. Increased urine sugar excretion is principally due to galactosuria and not glycosuria. There seems to be little or no impairment in glucose handling by the proximal tubule. Galactosemia should be suspected whenever there is a urinary reducing substance that does not react in a glucose oxidase test. The diagnosis can be confirmed by demonstration of deficient transferase activity in red blood cells, fibroblasts, leukocytes, or hepatocytes. Most infants with galactosemia are found through newborn metabolic screening.

Treatment

Galactosemia is treated by elimination of galactose from the diet. Acute symptoms and signs resolve in a few days. Cataracts will also regress to some extent. Even with early elimination of galactose, developmental delay, speech impairment, ovarian dysfunction, and growth retardation are common. Profound intellectual deficits are rare even in infants treated late.

Hereditary Fructose Intolerance

Etiology and Pathogenesis

Hereditary fructose intolerance is another disorder of carbohydrate metabolism associated with Fanconi syndrome.[10] Fructose intolerance is inherited as an autosomal recessive trait, with an estimated incidence of 1 in 20,000 live births. It is caused by a deficiency of the B isoform of the enzyme fructose-1-phosphate aldolase, which cleaves

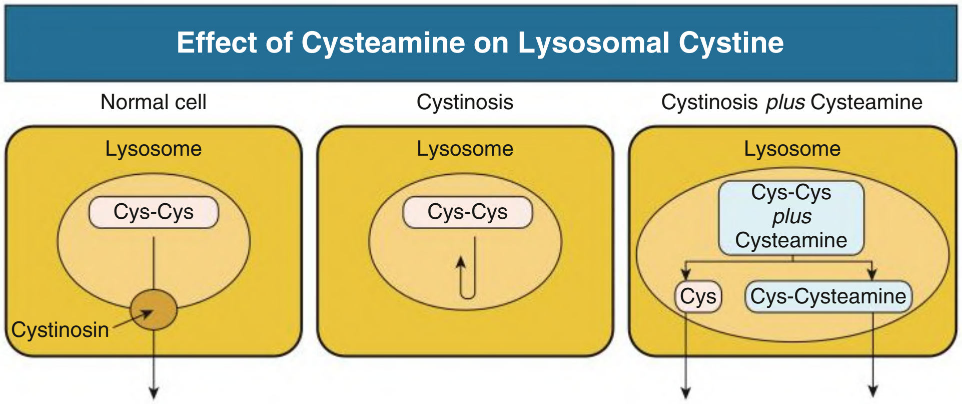

Fig. 50.6 Effect of Cysteamine on Lysosomal Cystine. In cystinosis, the transporter (cystinosin) for cystine (Cys-Cys) egress from the lysosome is defective, and cysteine accumulates. Cysteamine can easily enter the lysosome and combine with cystine, forming cysteine (Cys) and the mixed disulfide cysteamine-cysteine. Both these compounds can exit the lysosome through a transporter other than the cystine carrier.

fructose-1-phosphate into D-glyceraldehyde and dihydroxyacetone phosphate. Deficient activity of aldolase B leads to tissue accumulation of fructose-1-phosphate and reduced levels of adenosine triphosphate (ATP). Experimentally, mice with aldolase B deficiency can be rescued by blocking fructokinase, which prevents the accumulation of fructose-1-phosphate and maintains ATP.

Clinical Manifestations

Symptoms of hereditary fructose intolerance appear at weaning when fruit, vegetables, and sweetened cereals that contain fructose, sucrose, or sorbitol (the latter is converted to fructose in the body) are introduced. Affected children experience nausea, vomiting, and symptoms of hypoglycemia shortly after ingestion of fructose, sucrose, or sorbitol. These symptoms may progress to seizures, coma, and even death, depending on the amount consumed. When they are exposed to fructose, infants may have a catastrophic illness, with severe dehydration, shock, acute liver impairment, bleeding, and acute kidney injury (AKI). Concomitant serum biochemical findings after fructose ingestion are decreased glucose, phosphate, and bicarbonate and increased uric acid and lactic acid. Chronic exposure to fructose leads to failure to thrive, hepatomegaly and fatty liver, jaundice, hepatic cirrhosis, Fanconi syndrome, and nephrocalcinosis. Children with hereditary fructose intolerance quickly develop an aversion to sweets.

Diagnosis

The diagnosis should be suspected when symptoms develop after the ingestion of fructose. This can be confirmed by assaying the activity of fructose-1-phosphate aldolase in a liver biopsy specimen or through genetic testing of the aldolase B gene.

Treatment

Treatment of hereditary fructose intolerance involves strict avoidance of foods containing fructose and sucrose, but because most patients develop a strong aversion to such foods, this is usually easy. The greatest risk occurs during infancy, before those affected learn to avoid fructose.

Glycogenosis

Most patients with glycogen storage disease and Fanconi syndrome have an autosomal recessive disorder characterized by heavy glycosuria and increased glycogen storage in the liver and kidney, known as the *Fanconi-Bickel syndrome,* or *glucose-losing syndrome,* because the glucose losses can be massive.[11] These children have massive hepatomegaly, fasting hypoglycemia, and severe growth retardation in addition to Fanconi syndrome. The defect is deficient activity of the glucose transporter GLUT2 (Fig. 50.7). GLUT2 facilitates glucose exit from the basolateral side of the proximal tubule and intestinal cell and sugar entry and exit from the hepatocyte and pancreatic β cell. A few patients with type I glycogen storage disease have mild Fanconi syndrome but not Fanconi-Bickel syndrome. Treatment of this disorder is directed at the renal solute losses, treatment of rickets (which can be severe), and frequent feeding to prevent ketosis. Uncooked cornstarch has been shown to lessen the hypoglycemia and to improve growth.

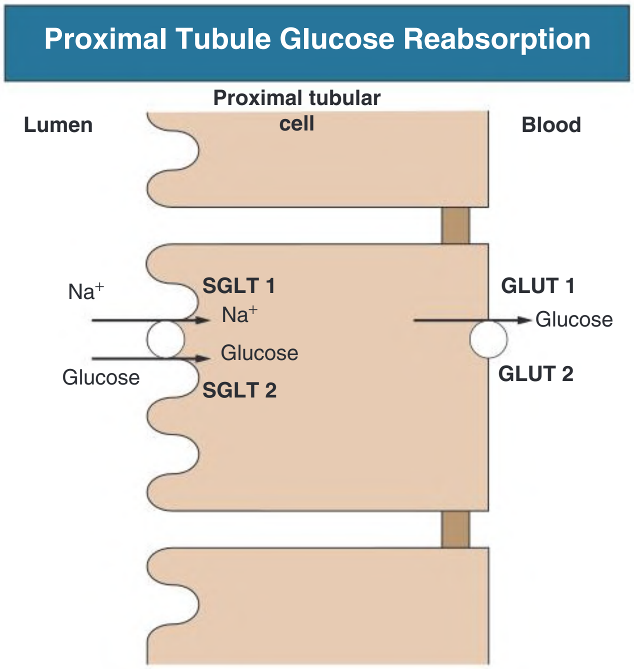

Fig. 50.7 Proximal Tubule Glucose Reabsorption. Glucose enters the proximal tubule cell coupled to Na^+ reabsorption from the lumen through a high-capacity, low-affinity transporter (SGLT2) in the early proximal tubule and a low-capacity, high-affinity transporter (SGLT1) in the late proximal tubule. Glucose exits the cell through the transporters (GLUT1 and GLUT2) located in the late and early proximal tubule, respectively. *GLUT1, -2,* Glucose transporter protein types 1, 2; *SGLT1,* sodium-glucose linked cotransporter type 1.

Tyrosinemia

Definition

Hereditary tyrosinemia type I, also known as *hepatorenal tyrosinemia,* is a defect of tyrosine metabolism affecting the liver, kidneys, and peripheral nerves.[12]

Etiology and Pathogenesis

The cause of hereditary tyrosinemia type I is a deficiency of fumarylacetoacetate hydrolase (FAH) activity; it is an autosomal recessive disorder. Decreased or absent FAH activity leads to accumulation of maleylacetoacetate (MAA) and fumarylacetoacetate (FAA) in affected tissues. These compounds can react with free sulfhydryl groups, reduce intracellular levels of glutathione, and act as alkylating agents. Maleylacetoacetate and fumarylacetoacetate are not detectable in plasma or urine but are converted to succinylacetoacetate and succinylacetone. Succinylacetone is structurally similar to maleic acid, a compound that causes Fanconi syndrome experimentally in rats and may be the cause of Fanconi syndrome in humans affected with tyrosinemia.

Clinical Manifestations

The liver is the major organ affected, evident as early as the first month of life. Such infants usually have severe disease and die in the first year. All children with tyrosinemia will eventually develop macronodular cirrhosis, and many develop hepatocellular carcinoma. Acute, painful peripheral neuropathy and autonomic dysfunction also can occur. Proximal renal tubular dysfunction is evident in all patients with tyrosinemia, especially those presenting after infancy. Nephromegaly is very common, and nephrocalcinosis may be seen. Glomerulosclerosis and impaired GFR may be seen with time.

Diagnosis

The diagnosis should be suspected with elevated plasma tyrosine and methionine levels together with their *p*-hydroxy metabolites. The presence of succinylacetone in blood or urine is diagnostic of hereditary tyrosinemia type I.

Treatment

The institution of a diet low in phenylalanine and tyrosine dramatically improves the renal tubular dysfunction. Nitisinone, which inhibits the formation of MAA and FAA, dramatically improves the renal and hepatic dysfunction.[12] Liver transplantation has been successfully used to treat patients with severe liver failure and prevent the development of hepatocellular carcinoma. Liver transplantation leads to rapid correction of Fanconi syndrome.

Wilson Disease

Definition

Wilson disease is an inherited disorder of copper metabolism that affects numerous organ systems.[13,14] It has an overall incidence of 1 in 30,000 live births. About 40% of patients present with liver disease, 40% with extrapyramidal symptoms, and 20% with psychiatric or behavioral abnormalities.

Etiology and Pathogenesis

Wilson disease is caused by a defect in the P-type copper–transporting adenosine triphosphatase ATP7B, which is highly expressed in the liver, kidney, and placenta. It impairs biliary copper excretion and the incorporation of copper into ceruloplasmin. These abnormalities cause excessive intracellular accumulation of copper in the liver, with subsequent overflow into other tissues, such as the brain, cornea, and renal proximal tubule.

Clinical Manifestations

Patients typically present with chronic liver disease, often with relatively high serum bilirubin relative to alkaline phosphatase (bilirubin [mg/dL]/alkaline phosphatase >4). Excessive storage of copper in the kidney leads to renal tubular dysfunction in most patients and full-blown Fanconi syndrome in some. Hematuria also has been noted. Renal plasma flow and GFR decrease as the disease progresses, but death from extrarenal causes occurs before the onset of kidney failure. Fanconi syndrome usually appears before the onset of hepatic failure. Hypercalciuria with development of kidney stones and nephrocalcinosis also have been reported. Besides proximal tubular dysfunction, abnormalities in distal tubular function, decreased concentrating ability, and distal RTA (type 1 RTA) also have been observed. Neurologic abnormalities, such as dysarthria and gait disturbances, may be the presenting symptom in young adults with Wilson disease. Kayser-Fleischer rings, dense brown copper deposits around the iris, may be visible but typically can be seen only with a slit lamp.

Kidney Pathology

Histologic examination of the kidney in untreated Wilson disease shows either no alteration on light microscopy or only some flattened proximal tubular cells without recognizable brush borders. Electron microscopy shows loss of the brush border, disruption of the apical tubular network, electron-dense bodies probably representing metalloproteins in the subapical region of tubule cell cytoplasm, and cavitation of the mitochondria with disruption of the normal cristae pattern. Rubeanic acid staining shows intracytoplasmic copper granules. The copper content of kidney tissue is greatly elevated.

Diagnosis

The diagnosis of Wilson disease should be suspected in children and young adults with unexplained neurologic disease, chronic active hepatitis, acute hemolytic crisis, behavioral or psychiatric disturbances, or the appearance of Fanconi syndrome. In such patients, the presence of Kayser-Fleischer rings is an important clue in making the diagnosis. Serum ceruloplasmin levels are decreased in 96% of patients with Wilson disease. A greatly increased urinary copper level is also useful in making the diagnosis, especially if it increases significantly with D-penicillamine. Liver copper levels are increased in untreated patients. Mutational analysis is also available.

Treatment

Treatment with penicillamine 1 to 1.5 g/day reverses the kidney dysfunction and potentially the hepatic and neurologic disease, depending on the degree of damage before the onset of therapy. Recovery, however, is quite slow. Trientine also can chelate copper and is indicated in patients who cannot tolerate penicillamine. Tetrathiomolybdate is a potent agent in removing copper from the body and has been used in some patients with neurologic disease to prevent the immediate worsening of symptoms that can occur with penicillamine. Zinc salts, which induce intestinal metallothionein and blockade of intestinal absorption of copper, are useful in maintenance therapy. Liver transplantation has been successful in some patients but should be reserved for those with liver failure.

Lowe Syndrome

Lowe syndrome (oculocerebrorenal syndrome) is characterized by congenital cataracts and glaucoma, severe developmental delay, hypotonia with diminished to absent reflexes, and renal abnormalities.[15,16] Fanconi syndrome is followed by progressive kidney impairment. End-stage kidney disease usually does not occur until the third to fourth decade of life.

Lowe syndrome is transmitted as an X-linked recessive trait. Despite this inheritance pattern, Lowe syndrome has occurred in a few females. The defective gene codes for phosphatidyl inositol 4,5-bisphosphate 5-phosphatase, *OCRL1,* involved with cell trafficking and signaling.

Light microscopy of the kidney is normal early in the disorder, with endothelial cell swelling and thickening and splitting of the glomerular basement membrane seen by EM. In the proximal tubule cells, there is shortening of the brush border and enlargement of the mitochondria, with distortion and loss of the cristae. Only symptomatic treatment is available.

Dent Disease

Definition

Dent disease is an X-linked recessive disorder characterized by low-molecular-weight proteinuria, hypercalciuria, nephrolithiasis, nephrocalcinosis, and rickets.[17,18] Affected males often have aminoaciduria, phosphaturia, and glycosuria. Kidney failure is common and may occur by late childhood. Hemizygous females usually have only proteinuria and mild hypercalciuria. X-linked recessive nephrolithiasis and X-linked recessive hypophosphatemic rickets have similar features, and most have a defect in the renal ClC-5 chloride channel. Dent disease type 2 is a clinically similar disease affecting males, but there is a mutation in the same gene that causes Lowe syndrome, although patients with Dent type 2 do not have the brain or eye involvement seen in Lowe syndrome.[17]

Etiology and Pathogenesis

Most of these disorders are caused by a mutation in the *CLCN5* gene leading to inactive ClC-5 chloride channel function (see Fig. 50.2). The ClC-5 chloride channel spans the membrane of preendocytic vesicles just below the brush border of the proximal tubule. This channel plays a role in the acidification of these vesicles by a proton pump. Lack of this Cl^- channel interferes with protein reabsorption from the tubule through the megalin-cubilin receptor system and cell surface receptor recycling, which may explain the phosphaturia, glycosuria, and aminoaciduria.

The defective *OCRL1* in patients with Dent disease type 2 interferes with normal cell protein trafficking. The kidney disease is similar to that seen in Dent disease type 1. Although patients do not have the eye and brain disease seen in patients with Lowe, a few patients with Dent type 2 have a mild intellectual deficit, hypotonia, and subclinical cataracts.

Filtered PTH is also reabsorbed by the megalin-cubilin system for degradation in the lysosome. Decreased PTH reabsorption allows increased binding to luminal PTH receptors and increased endocytosis of luminal phosphate transporters, leading to increased phosphaturia.[1]

Mitochondrial Cytopathies

Definition

Mitochondrial cytopathies are a diverse group of diseases with abnormalities in mitochondrial DNA that lead to mitochondrial dysfunction in various tissues.[19]

Clinical Manifestations

Most of the mitochondrial cytopathies manifest with neurologic disorders such as myopathy, myoclonus, ataxia, seizures, external ophthalmoplegia, stroke-like episodes, and optic neuropathy. Other manifestations include retinitis pigmentosa, diabetes mellitus, exocrine pancreatic insufficiency, sideroblastic anemia, sensorineural hearing loss, pseudoobstruction of the colon, hepatic disease, cardiac conduction disorders, and cardiomyopathy.

The most common kidney manifestation associated with mitochondrial cytopathies is Fanconi syndrome, although a number of patients have been described with focal segmental glomerulosclerosis (FSGS) and corticosteroid-resistant nephrotic syndrome. All the patients with kidney abnormalities have had extrarenal disorders, mainly neurologic disease. Most patients present in the first months of life and die soon afterward.

Diagnosis

A clue to mitochondrial cytopathies is elevated serum or cerebrospinal fluid lactate levels, especially in association with an altered lactate-to-pyruvate ratio, suggesting a defect in mitochondrial respiration. The presence of "ragged red fibers," a manifestation of abnormal mitochondria, in a muscle biopsy specimen is another clue, especially with large abnormal mitochondria on EM of muscle tissue.

Treatment

There is little to offer these patients in terms of definitive therapy. Low mitochondrial enzyme complex III activity can be treated with menadione or ubidecarenone. Deficient mitochondrial enzyme complex I activity may be treated with riboflavin and ubidecarenone. Ascorbic acid has been used to minimize oxygen free radical injury. High-lipid, low-carbohydrate diet has been tried in cytochrome c oxidase deficiency.

Idiopathic Fanconi Syndrome

A number of patients develop the complete Fanconi syndrome in the absence of any known cause. Traditionally called *adult* Fanconi syndrome because it was thought that only adults were affected, it is now clear that children may be affected, and a more proper designation is *idiopathic* Fanconi syndrome. Not all the features of Fanconi syndrome may be present when the patient is first seen but do appear with time. Idiopathic Fanconi syndrome can be inherited in an autosomal dominant, autosomal recessive, or even X-linked pattern. However, most cases occur sporadically, with no evidence of genetic transmission. The prognosis is variable, and some patients develop ESKD 10 to 30 years after onset of symptoms. A few patients have undergone kidney transplantation; in some, Fanconi syndrome has recurred in the allograft without evidence of rejection, suggesting an extrarenal cause of the idiopathic form.

Renal morphologic descriptions of such cases are scanty. In some reports, no abnormalities were found, and, in others, tubular atrophy with interstitial fibrosis was interspersed with areas of tubular dilation. Greatly dilated proximal tubules with swollen epithelium and grossly enlarged mitochondria with displaced cristae also have been noted.

ACQUIRED CAUSES OF FANCONI SYNDROME

Numerous substances can injure the proximal renal tubule. Injury can range from an incomplete Fanconi syndrome to acute tubular necrosis (ATN) or ESKD. The extent of the tubular damage varies depending on the type of toxin, amount ingested, and host characteristics. A careful history of possible toxin exposure and recent medications is important in patients with tubular dysfunction. Box 50.1 lists the more common causes of acquired Fanconi syndrome.

Heavy Metal Intoxication

A major cause of proximal tubular dysfunction is acute heavy metal intoxication, principally lead and cadmium (see Chapter 65). In lead poisoning, renal tubular dysfunction, mainly aminoaciduria and mild glycosuria and phosphaturia, is usually overshadowed by the development of chronic kidney disease (CKD) and involvement of other organs, especially the central nervous system.[20] Fanconi syndrome associated with cadmium poisoning is associated with severe bone pain, giving rise to the name itai-itai ("ouch-ouch") disease for its occurrence in Japanese patients affected by industrial contamination of the soil.[21]

Tetracycline

Outdated tetracycline causes reversible Fanconi syndrome even in therapeutic doses. Recovery is rapid when the degraded drug is stopped. The compound responsible for the tubule dysfunction is anhydro-4-tetracycline, formed from tetracycline by heat, moisture, and low pH.

Cancer Chemotherapy Agents

A number of cancer chemotherapy agents have been associated with Fanconi syndrome and renal tubular dysfunction, especially cisplatin and ifosfamide. Carboplatin has been associated with reduced GFR and magnesium wasting but not Fanconi syndrome. The nephrotoxicity of both cisplatin and ifosfamide is dose dependent and often irreversible. Besides the usual manifestations of Fanconi syndrome, cisplatin toxicity is characterized by hypomagnesemia, caused by hypermagnesuria, which can be extremely severe, persistent, and difficult to treat.[22,23] Ifosfamide is more often associated with hypophosphatemic rickets.[22] Chloroacetaldehyde, a metabolite of ifosfamide, appears experimentally to cause Fanconi syndrome. Both ifosfamide and cisplatin can also cause an irreversible reduction in GFR.

Other Drugs and Toxins

Exposure to a wide range of toxins may give rise to Fanconi syndrome, often in association with a reduced GFR, including 6-mercaptopurine, toluene (glue sniffing), and Chinese herbal medicines containing *Aristolochia* spp (see Chapter 79).[24] In addition, anecdotal reports have associated Fanconi syndrome with valproic acid (valproate), suramin, gentamicin, and ranitidine. Antiviral medications, especially antiretroviral agents such as tenofovir, are an increasingly common cause of Fanconi syndrome.[25]

Dysproteinemias

Dysproteinemia[26] from multiple myeloma, light-chain proteinuria,[27] Sjögren syndrome, and amyloidosis is sometimes associated with Fanconi syndrome, which appears to be correlated with urinary free light chains that can cause proximal tubule dysfunction through intracellular crystallization or lysosomal dysfunction.[28]

Glomerular Disease

Nephrotic syndrome is rarely associated with Fanconi syndrome. Most of these patients have FSGS, and the occurrence of Fanconi syndrome heralds a poor prognosis.

After Acute Kidney Injury

Tubular dysfunction can occur transiently during recovery from AKI from any cause, regardless of whether a known tubular toxin was originally implicated.

After Kidney Transplantation

Fanconi syndrome has appeared rarely after kidney transplantation. The pathogenesis probably is multifactorial, including sequelae of ATN, rejection, nephrotoxic drugs, ischemia from renal artery stenosis, and residual hyperparathyroidism.

FAMILIAL GLUCOSE-GALACTOSE MALABSORPTION AND HEREDITARY RENAL GLYCOSURIA

Definition

Renal glycosuria refers to the appearance of readily detectable glucose in the urine when the serum glucose concentration is in a normal range (see Fig. 50.3). When the serum glucose concentration is in a physiologic range, virtually all the filtered glucose is reabsorbed in the proximal tubule.[29] Filtered glucose enters the proximal tubule through two specific carriers (SGLT1 and SGLT2) coupled to sodium and exits the cell through the sugar transporters GLUT1 and GLUT2 (see Fig. 50.7). However, when the serum level exceeds the physiologic range, the filtered load exceeds the capacity of these carriers, and glucose begins to appear in the urine; this is termed the *renal threshold.*

Etiology and Pathogenesis

Familial glucose-galactose malabsorption is a rare autosomal disorder caused by mutations in the gene coding for the brush border sodium-glucose cotransporter SGLT1, which is found in the intestinal cell and the S_3 segment of the proximal renal tubule cell. The disorder is characterized by the neonatal onset of life-threatening diarrhea from the intestinal malabsorption of glucose and galactose, which resolves rapidly with the removal of glucose and galactose and its dipeptide, lactose, from the diet. These patients frequently also have mild renal glycosuria.

Familial renal glycosuria occurs with an incidence of 1 in 20,000 live births and can be inherited as a heterozygous, homozygous, or mixed heterozygous trait.[29] This disorder is caused by mutations in the *SLC5A2* gene that codes for the SGLT2 glucose transporter found in the early portion of the proximal tubule. Inhibitors of SGLT2, which mimic this genetic defect, have been used recently in type 2 diabetes to lower hyperglycemia without causing weight gain or aggravating hyperinsulinism (see Chapter 32).

In the past, renal glycosuria was divided into three types based on the reabsorption patterns observed during glucose infusion studies, but this typing system has been questioned because clearance data suggest patients with renal glycosuria have rates of glucose reabsorption that vary from virtually no reabsorption to near-normal rates, rather than three distinct types, reflecting different mutations in the *SLC5A2* gene and differing amounts of functional protein.

Natural History

Patients with familial glucose-galactose malabsorption appear to grow and develop normally with removal of the offending sugars from the diet. The clinical course of hereditary renal glycosuria is benign, except for a few patients with polyuria and salt wasting, and it is not a precursor to diabetes mellitus. Patients need to be aware of the condition in order not to receive unnecessary diagnostic investigations or even treatment for presumed diabetes mellitus.

AMINOACIDURIAS

As with glucose, amino acids are almost completely reabsorbed in the proximal tubule by a series of specific carriers. Studies have described a number of inherited disorders resulting in the incomplete reabsorption of a specific amino acid or a group of amino acids[30] (Table 50.2). Most do not result in kidney disease.

Cystinuria

Definition

Cystinuria is characterized by the excessive urinary excretion of cystine and the dibasic amino acids ornithine, lysine, and arginine and accounts for about 1% to 2% of all kidney stones and 6% to 8% of pediatric kidney stones.[31,32]

Etiology and Pathogenesis

These four amino acids share a transport system on the brush border membrane of the proximal tubule. Because of the relative insolubility of cystine when its urine concentration exceeds 250 mg/L (1 mmol/L), patients with cystinuria have recurrent kidney stones.

Cystinuria is an autosomal recessive trait with a disease incidence of 1 in 15,000 births.[31,32] Early studies suggested there were three genetic types on the basis of in vitro studies of intestinal transport and amino acid excretion in heterozygotes. More recently, two genes (*SLC3A1* coding for the protein rBAT and *SLC7A9* for the protein $b^{0,+}$AT) have been identified that are defective in cystinuria. *SLC3A1* heterozygotes have normal excretion rates for cystine. *SLC7A9* heterozygotes have cystine excretion rates that range from normal to almost that of homozygous patients. Based on these data, a newer classification proposed type A for mutations in both *SLC3A1* genes and type B for mutations in *SLC7A9*.[31,32] Type AB is a compound heterozygote. Type A accounts for 38% of cystinuria patients, type B for 47%, and type AB for 14%.

Clinical Manifestations

Cystine stones (calculi) are typically yellow-brown (Fig. 50.8A) and radiopaque (see Fig. 50.8B). Cystine crystals appear as microscopic, flat hexagons in the urine (see Fig. 50.8C), and this is a clue to the diagnosis.

Diagnosis

Patients can be tested for cystinuria with the cyanide-nitroprusside test, but type B heterozygotes also may give a positive result. The definitive test is to quantify cystine and dibasic amino acid excretion in a 24-hour urine specimen. Homozygotes excrete more than 118 mmol cystine per mmol creatinine (250 mg/g creatinine). Genetic testing for mutations in the *SLC3A1* and *SLC7A9* is also readily available.

Treatment

The aim of therapy in cystinuria is to lower the urine cystine concentration to less than 250 mg/L (1 mmol/L).[31,32] The first step is to increase daily fluid intake to greater than 2 L/1.73 m^2 and restrict sodium intake. However, because most patients with cystinuria excrete 0.5 to 1 g/day of cystine, a urine output of 2 to 4 L/day is needed to achieve this goal. Cystine solubility increases in alkaline urine, but the urine pH must be > 7 to 7.5 to be effective. In patients with recurrent stone disease, thiols such as penicillamine and tiopronin are extremely useful through the formation of a more soluble, mixed disulfide of the thiol and cysteine from cystine. Tiopronin is the most commonly prescribed thiol because it has fewer side effects than penicillamine and is started at a low dose (100 mg bid) and slowly increased (doses >1000 mg/day have little added effect) to achieve a urine cystine concentration below 250 mg/L in conjunction with a high fluid intake. Penicillamine is also effective, and the dose should be slowly increased to minimize side effects. Captopril can be useful (an effect resulting from its thiol structure, not its angiotensin-converting enzyme inhibitor effect), but the dose range (75–150 mg/day) may be limited by its hypotensive effects.

TABLE 50.2 Inherited Aminoacidurias

Disease	Clinical Findings	Urine Amino Acids
Cystinuria	Urolithiasis	Cystine, lysine, ornithine, arginine
Hartnup disease	Rash, neurologic disease	Neutral amino acids
Iminoglycinuria	None	Proline, hydroxyproline glycine
Lysinuric protein intolerance	Hyperammonemia, vomiting, diarrhea	Dibasic amino acids

HEREDITARY DEFECTS IN URIC ACID HANDLING

Hereditary Renal Hypouricemia

Hereditary renal hypouricemia is a rare autosomal recessive disorder characterized by very low serum uric acid levels (<2.5 mg/dL [<150 μmol/L] in adult men and <2.1 mg/dL [<124 μmol/L] in adult women) and increased uric acid clearance, ranging from 30% to 150% of the filtered load.[33] In the normal kidney, uric acid is both reabsorbed and secreted in the proximal tubule by two different uric acid–anion exchange transporters and a voltage-sensitive pathway. In some patients, the defect is in the gene *SLC22A12* that codes for the protein URAT1; other patients have been found to have mutations in *SLC2A9 (GLUT9)*. Most patients do not have symptoms, and hypouricemia is found incidentally when a low serum uric acid concentration is noted during routine serum chemistry evaluation. About one-fourth of patients with renal hypouricemia have had kidney stones, but only one-third of these were uric acid stones. There also may be hypercalciuria, and a few patients have had exercise-induced AKI, thought to be caused by acute tubular injury by passage of uric acid "gravel" in association with volume depletion and reduced urine pH. Most patients require no treatment, but if forming uric acid stones, they should maintain a high fluid intake. Urine alkalinization and allopurinol can be used for patients with persistent uric acid stones.

Fig. 50.8 Cystinuria. (A) Rough and smooth cystine calculi. (B) Plain radiograph of a cystine calculus in the right renal pelvis and further multiple parenchymal calculi. (C) Urine microscopy showing characteristic flat hexagonal crystals.

Autosomal Dominant Tubulointerstitial Kidney Disease

Autosomal dominant tubulointerstitial kidney disease (ADTKD) is a group of rare disorders characterized by progressive loss of kidney function, bland urinary sediment, and autosomal dominant inheritance.[34] These diseases were previously classified as familial juvenile hyperuricemic nephropathy (FJHN) or medullary cystic kidney disease type 2 (MCKD2). However, as neither hyperuricemia nor renal cysts are pathognomonic, these disorders have been reclassified as ADTKD. There are four main genes that are known to cause ADTKD: *UMOD* (uromodulin), *MUC1* (mucin-1), *REN* (renin), and HNF1B (hepatocyte nuclear factor β).

The *UMOD* gene encodes for the Tamm-Horsfall/uromodulin protein, which is synthesized in the epithelial cells of the thick ascending limb (TAL) of the loop of Henle. Defective protein is retained in the endoplasmic reticulum and probably leads to inflammation, interstitial fibrosis, and functional abnormalities of the TAL. The defective protein also causes decreased salt and water reabsorption because *UMOD* plays a role in regulating the sodium-potassium-chloride transporter[35] and the rat outer medullary potassium channel.[36] The decreased salt and water reabsorption leads to increased proximal salt reabsorption and secondarily of uric acid, leading to hyperuricemia.

The *MUC1* gene encodes for mucin-1, a transmembrane protein, which is expressed in the loop of Henle, distal tubule, and collecting duct. The pathogenesis of how defective mucin-1 leads to tubulointerstitial fibrosis remains unclear. Patients with ADTKD-*MUC1* can develop hyperuricemia as well, but gout tends to be a later manifestation of the disease.

The *REN* gene encodes for preprorenin, which is then converted to prorenin and subsequently renin. Mutation in this gene leads to retention of preprorenin, prorenin, or renin protein in the endoplasm, depending on the location of the defect. As a result of a low renin state, patients with ADTKD-*REN* are at risk of mild hypotension, mild hyperkalemia, increased risk of AKI in the setting of volume depletion, and transient childhood anemia. Hyperuricemia and gout may also occur, though the exact mechanism is unclear.

The *HNF1B* gene encodes hepatocyte nuclear factor β, a transcription factor that regulates multiple genes located in the kidney, pancreas, and liver. Patients with *HNF1B* mutations have variable presentations and can present with extrarenal manifestations. The disease ADTKD-*HNF1B* occurs in a small subset of patients with *HNF1B* mutations and is characterized by kidney cysts, maturity onset diabetes of youth, genital abnormalities, pancreatic atrophy, and hypomagnesemia.

Diagnosis of ADTKD should be suspected in patients with CKD and family history of autosomal dominantly inherited CKD. In patients with ADTKD-*UMOD*, the diagnosis is suggested by a fractional excretion of uric acid of less than 5% (normal, 10%–15%). Definitive diagnosis can be made by genetic testing of the *UMOD, MUC1, REN* (renin), and HNF1B genes. There are no specific treatments for ADTKD, and management is largely based on CKD care (see Chapter 82). Controversy exists as to whether lowering of serum uric acid slows the progression of CKD; the studies reporting benefit have usually involved starting a xanthine oxidase inhibitor early in the disease.

More recently from genome-wide association studies, variants in the *UMOD* gene have been identified as risk factors for CKD and hypertension.[37] The variants are common, are in the noncoding regions of the *UMOD* gene, and lead to an increase in functional UMOD protein in contrast to the mutation in ADTKD.

SELF-ASSESSMENT QUESTIONS

A 3-year-old boy presents with failure to thrive and photophobia. His urine contains glucose and +1 protein with a pH of 5.5. Serum chemistries are as follows:

Na^+ 135 mmol/L
K^+ 2.5 mmol/L
Cl^- 111 mmol/L
HCO_3^- 15 mmol/L
Glucose 91 mg/dL (5 mmol/L)
Phosphorus 2.5 mg/dL (0.8 mmol/L) (normal for age, 4.5–5.5 mg/dL)
Creatinine 0.4 mg/dL (35 μmol/L)

1. Slit-lamp examination of his eyes shows tinsel-like refractile opacities in his cornea. What is the most likely diagnosis?
   A. Lowe syndrome
   B. Cystinosis
   C. Tyrosinemia
   D. Hereditary fructose intolerance
   E. Galactosemia
2. Cystinuria is associated with which amino acids in the urine besides cystine?
   A. Arginine, lysine, ornithine
   B. Arginine, lysine, histidine
   C. Lysine, ornithine, histidine
   D. Ornithine, glycine, alanine
   E. Ornithine, glycine, serine
3. What is the most common cause of Fanconi syndrome in adults?
   A. Multiple myeloma
   B. Medications
   C. Nephrotic syndrome
   D. Cystinosis
   E. Cadmium
4. The defect in familial glycosuria is an abnormality in which transporter?
   A. GLUT1
   B. ClC-5
   C. Na^+-K^+-$2Cl^-$
   D. Na^+/H^+
   E. SLGT2
5. Most patients with Dent disease have a mutation in which transporter?
   A. Na^+-K^+-$2Cl^-$
   B. GLUT2
   C. Na^+/H^+
   D. ClC-5
   E. Na^+-Cl^-

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